What two numbers mult to give $A \cdot C = 6(-12) = -72$, and add to B, (-1). $-9 \cdot 8 = -72$, also $-9 + 8 = 1$

need to find 4th term

ex. Factor a trinomial by grouping: $6x^2 - x - 12$ $\overset{A}{}\ \overset{B}{}\ \overset{C}{}$ $=$

$6x^2 - 9x + 8x - 12 =$

$(6x^2 - 9x) + (8x - 12) =$

$3x(2x - 3) + 4(2x - 3) =$

$(2x - 3)(3x + 4)$ FOIL to check.

Practice.

1) Factor completely:

$(x-4)^3(4)(3x+1)^3(3) + (3x+1)^4(3)(x-4)^2$

2) Factor completely: $2y^4 - 8y^2$

3) Find the product $(m+n-3)(m-n-3)$

4) Find the product $(x+y)(x-y)(x^2+y^2)$

5) Find the sum $(5+2\sqrt{6})^2 + (5-2\sqrt{6})^2$

Answer. 1) $3(x-4)^2(3x+1)^3(7x-15)$ 2) $2y^2(y-2)(y+2)$

3) $m^2 - 6m + 9 - n^2$ 4) $x^4 - y^4$ 5) 98

Lecture 4: Section A.4
Rational Expressions

Domain

Simplifying rational expressions

Multiplying and dividing rational expressions

Adding and subtracting rational expressions

Complex fractions

Expressions with negative exponents

Difference quotient

F12 4th CAR
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Lecture 4: Section A.4
Rational Expressions

$\sqrt{-25}$ is not real
$= 5i$

Def. The **domain** of an expression is the set of real numbers for which the expression is defined.

Look to see what makes it undefined.

ex. Find the domain.

① can't ÷ by 0
② can't have neg under an even root.

1) $2x^5 - 6x - 1$

D: all reals or $(-\infty, \infty)$

2) $\sqrt{x+3}$

$x+3 \geq 0$
$x \geq -3$ or $[-3, \infty)$

3) $\dfrac{x-1}{x+5}$

All reals, except $x \neq -5$.

$x+5 \neq 0$
$x \neq -5$

$(-\infty, -5) \cup (-5, \infty)$

Def. A **rational expression** is the quotient of two polynomials.

Simplifying Rational Expressions

A fraction is in **simplest form** if its numerator and denominator have no common factors.

Use the Cancellation Property: $\frac{a \cdot \cancel{c}}{b \cdot \cancel{c}} = \frac{a}{b}$ if $c \neq 0$

ex. Simplify $\frac{x^3 - 4x}{x^2 - x - 2}$ and find its domain. Hint: First find the domain, then simplify.

D: $x^2 - x - 2 \neq 0$

$(x-2)(x+1)$

thus $x \neq 2, -1$ domain

$$\frac{x(x^2-4)}{(x-2)(x+1)} = \frac{x(x+2)\cancel{(x-2)}}{\cancel{(x-2)}(x+1)}$$

$$= \boxed{\frac{x(x+2)}{x+1},\ x \neq 2}$$

Note: For expressions to be equivalent, they must have the same domain.

Multiplying and Dividing Rational Expressions

Recall: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

1st do domain: $x \neq -4, 1$

ex. $\dfrac{x^2+2x-3}{x^2+8x+16} \cdot \dfrac{3x+12}{x-1} = \dfrac{(x+3)\cancel{(x-1)}}{(x+4)^{\cancel{2}}} \cdot \dfrac{3\cancel{(x+4)}}{\cancel{x-1}} =$

$= \boxed{\dfrac{3(x+3)}{x+4}, \; x \neq 1}$

Domain: $x \neq \pm 2, -3, 4, -1$

Note: $4-x = -1(-4+x) = -(x-4)$

tricky

ex. $\dfrac{4-x}{x^4-16} \div \dfrac{x^2-3x-4}{x^2+5x+6} = \cancel{\dfrac{4-x}{(x+2)(x-2)(x^2+4)}}$

$(x^2-4)(x^2+4)$

$-(x-4)$

$\dfrac{4-x}{(x+2)(x-2)(x^2+4)} \div \dfrac{(x-4)(x+1)}{(x+2)(x+3)} = \dfrac{-\cancel{(x-4)}}{\cancel{(x+2)}(x-2)(x^2+4)} \cdot \dfrac{\cancel{(x+2)}(x+3)}{\cancel{(x-4)}(x+1)}$

$= \boxed{\dfrac{-(x+3)}{(x-2)(x^2+4)(x+1)}, \; x \neq -2, -3, 4, 2, -1}$

Adding and Subtracting Rational Expressions

Least Common Denominator (LCD) is the least common multiple (LCM) of the denominators. To find it:

1. Factor each denominator completely.
2. The LCD is the product of each prime factor of the denominators, with each factor raised to its highest power in any denominator.

To add or subtract, write each rational expression with LCD as its denominator, and use

$$\frac{a}{b} \pm \frac{c}{b} = \frac{a \pm c}{b}, \; b \neq 0 \qquad LCD = 6x^2y^3$$

<u>ex.</u> Perform the operations and simplify:

1) $\frac{2}{3}x^{-2}y^{-1} + \frac{5}{6}x^{-1}y^{-3} = \frac{2}{3x^2y} + \frac{5}{6xy^3} =$

$$\frac{2}{3x^2y}\left(\frac{2y^2}{2y^2}\right) + \frac{5}{6xy^3}\left(\frac{x}{x}\right) = \frac{4y^2}{6x^2y^3} + \frac{5x}{6x^2y^3} =$$

$$\boxed{\frac{4y^2+5x}{6x^2y^3}}$$

LCD: $(x+1)^2(x-1)$

$(x+1)(x-1)$

2) $\dfrac{1}{x+1} - \dfrac{2}{(x+1)^2} + \dfrac{3}{x^2-1}$

$$\left(\frac{1}{x+1}\right)\left(\frac{(x+1)(x-1)}{(x+1)(x-1)}\right) - \frac{2}{(x+1)^2}\left(\frac{x-1}{x-1}\right) + \left(\frac{3}{x^2-1}\right)\left(\frac{x+1}{x+1}\right) =$$

$$\frac{(x+1)(x-1) - 2(x-1) + 3(x+1)}{(x+1)^2(x-1)} = \frac{x^2-1-2x+2+3x+3}{(x+1)^2(x-1)}$$

$$= \boxed{\frac{x^2+x+4}{(x+1)^2(x-1)}}$$

Complex Fractions

<u>Def.</u> A **complex fraction** is a quotient containing rational expressions.

To Simplify complex fractions:

Method 1. Perform operations in numerator and denominator to write each as a single fraction, then divide by inverting the denominator and multiply.

<u>ex.</u> Simplify: $\dfrac{2-\dfrac{3}{y}}{4-\dfrac{9}{y^2}} = \dfrac{2\left(\frac{y}{y}\right)-\frac{3}{y}}{4\left(\frac{y^2}{y^2}\right)-\frac{9}{y^2}} = \dfrac{\frac{2y-3}{y}}{\frac{4y^2-9}{y^2}} =$

$$\frac{2y-3}{y} \div \frac{4y^2-9}{y^2} = \frac{2y-3}{\cancel{y}} \cdot \frac{y^{\cancel{2}}}{4y^2-9} =$$

$$= \frac{y\cancel{(2y-3)}}{(2y+3)\cancel{(2y-3)}} = \boxed{\frac{y}{2y+3},\ y \neq \frac{3}{2}, 0}$$

LCM: $(x+3)(x-3)$

Method 2. Multiply numerator and denominator by the LCM of all denominators.

ex. Simplify: 1) $\left(\dfrac{\dfrac{x+6}{x^2-9}}{\dfrac{x}{x-3}-\dfrac{2}{x+3}}\right)\left(\dfrac{\dfrac{(x+3)(x-3)}{1}}{\dfrac{(x+3)(x-3)}{1}}\right) =$

$$\frac{x+6}{x(x+3)-2(x-3)} = \frac{x+6}{x^2+3x-2x+6} = \boxed{\frac{x+6}{x^2+x+6},\ x \neq \pm 3}$$

2) $\left(\dfrac{\dfrac{x^2}{\sqrt{x^2+1}}-\sqrt{x^2+1}}{x^2}\right)\left(\dfrac{\dfrac{\sqrt{x^2+1}}{1}}{\dfrac{\sqrt{x^2+1}}{1}}\right) = \dfrac{x^2-(x^2+1)}{x^2\sqrt{x^2+1}}$

$$= \left(\frac{-1}{x^2\sqrt{x^2+1}}\right)\left(\frac{\sqrt{x^2+1}}{\sqrt{x^2+1}}\right) = \boxed{\frac{-\sqrt{x^2+1}}{x^2(x^2+1)}}$$

$\frac{1}{6} \cdot \frac{?}{1} = \frac{1}{3}$ $\frac{1}{6} \cdot \frac{?}{1} = \frac{1}{2}$ $-\frac{2}{3} + \frac{5}{3} = \frac{3}{3} = 1$

Expressions with Negative Exponents

Factor out the common factor with the **smaller** exponent.

NOTE: When factoring, you **subtract** the exponents.

For example: $2x^{-5/3} - 3x^{-2/3} = x^{-5/3}\left(2 - 3x^{-2/3 - -5/3}\right) =$

$x^{-5/3}(2 - 3x) = \boxed{\frac{2-3x}{x^{5/3}}}$

ex. Write $\frac{1}{3}x^{-2/3}(x+1)^{1/2} + \frac{1}{2}x^{1/3}(x+1)^{-1/2} =$

as a single quotient without negative exponents.

$\frac{1}{6} x^{-2/3}(x+1)^{-1/2}\left[2(x+1)^{1/2 - -1/2} + 3x^{1/3 - -2/3}\right] =$

$\frac{1}{6} x^{-2/3}(x+1)^{-1/2}\left(2(x+1) + 3x\right) =$

$$\boxed{\frac{5x+2}{6x^{2/3}(x+1)^{1/2}}}$$

Difference Quotient

<u>ex.</u> Rewrite the difference quotient $\left(\frac{\sqrt{4+h}-2}{h}\right)\left(\frac{\sqrt{4+h}+2}{\sqrt{4+h}+2}\right)$ by rationalizing its numerator.

$$= \frac{\cancel{4}+h-\cancel{4}}{h(\sqrt{4+h}+2)} = \frac{\cancel{h}}{\cancel{h}(\sqrt{4+h}+2)} = \boxed{\frac{1}{\sqrt{4+h}+2}}$$

$h \neq 0$

Practice.

1) Simplify $\dfrac{2x^2 - 9x + 4}{12 + x - x^2}$ and find its domain.

2) Factor $(x+1)^{-1} - (x+1)^{-3}(x+4)$ completely without negative exponents.

3) Write $\dfrac{(x^2+4)^{1/2} - x^2(x^2+4)^{-1/2}}{x^2+4}$ as a single quotient without negative exponents.

4) Let $a > 0$. If $a^{1/2} + a^{-1/2} = 3$, find (a) $a + a^{-1}$ and (b) $a^2 + a^{-2}$. Hint: Square both sides of the equation.

5) Let $x > 0$. If $x + \dfrac{1}{x} = 7$, find $\sqrt{x} + \dfrac{1}{\sqrt{x}}$.

Answer. 1) $\dfrac{1-2x}{x+3}$, D: $x \neq -3$ 2) $\dfrac{x^2+x-3}{(x+1)^3}$

3) $\dfrac{4}{(x^2+4)^{3/2}}$ 4) 7, 47 5) 3